

B Manoranjan Patra

Design Verification Engineer

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SUMMARY

Design Verification Engineer with around 3 years of expertise in pre-silicon verification using SystemVerilog/UVM. Proven experience in verifying complex protocols (PCIe Gen4 and AXI4) and debugging RTL designs. Adept at developing scalable testbenches, achieving functional coverage closure, and collaborating with cross-functional teams to deliver bug-free designs.

EDUCATION

B.E. in Electronics and Telecommunication Engineering

IGIT, Sarang | 2021 | 8.46 CGPA

Higher Secondary (HSC)

MP&EV English Medium School, Visakhapatnam | 2017 | 84.8%

Secondary School (SSC)

Kendriya Vidyalaya, Khurda Road | 2015 | 9.8 CGPA

WORK EXPERIENCE

Design Verification Engineer

SiyaCon Technologies | 2023 – Present

Developed System Verilog , C and UVM-based testbenches and verification environments.

Created components such as drivers, monitors, scoreboards, and functional coverage collectors.

Worked on RTL simulation, debugging, waveform analysis, and log verification using Cadence SimVision.

Collaborated with designers to identify bugs and provide fixes in pre-silicon verification stage.

Involved in spec review, testcase development, regression setup, and reporting. Worked on both IP and SoC verification.

PROJECTS

HUDSON (SoC FOR WEARABLES)

Worked on the verification of **OTP (One-Time Programmable memory)**, ensuring correct programming, readback, and security features.

Verified **PINMUX** functionality to validate accurate signal routing and multiplexing across different subsystems.

Handled **GPIO** controller verification, focusing on direction control, interrupt functionality, and interface behavior.

Developed and executed test cases and simulations to validate block-level and integration-level functionality.

Contributed to debugging and coverage analysis to identify corner cases and improve overall verification quality.

Ran regular regressions to monitor design stability, track failures, and ensure consistent verification progress.

PCI Express (Gen 4) VIP

Verified endpoint, root complex, switch, and dual-mode configurations at 16 GT/s.

Worked on physical layer analysis and DLLP packet handling.

Implemented TLP/Ordered Set packing/unpacking and LTSSM state machine analysis.

Simulated traffic patterns and analyzed waveform for protocol compliance.

Extracted configuration space test cases and link transaction test cases.

Debugged ordered set mismatches (TS1/TS2), trained receiver, and ensured link establishment.

Dispatch Engine Verification (IP)

Studied architecture specs and extracted verification features.

Developed UVM testbenches for DMA descriptor verification.

Implemented and debugged complex test scenarios for functional coverage.

Collaborated with architecture team to understand Reaper cluster interactions.

Ensured test consistency across multiple clusters using reusable sequences and configs.

AXI4 Protocol Verification

Designed UVM components (driver, monitor) for master/slave configuration.

Developed read/write sequences and monitored protocol transactions.

Analyzed waveform data and verified transfer phases and burst types.

Worked with 512-bit wide data path and separate DMA engines.

Implemented scoreboard logic to match expected vs. observed transactions.

TECHNICAL ADD ONs

Developed and executed multiple test cases for various projects to ensure functional correctness. Gained foundational understanding of Gate-Level Simulation (GLS) verification for digital designs. Debugged and resolved RTL and verification environment bugs to improve design robustness.

SKILLS

Languages: Verilog , SystemVerilog and python(Basics)

Methodology: UVM

Protocols: AXI4, AXI4-Lite, APB, PCIe Gen 4

EDA and Vplan Tools: Cadence SimVision , Vmanager ,Questasim,Synopsys VCS.

ACHIEVEMENTS

National level kho-kho (2014)

Had the opportunity to represent the KVS Foundation at the national level in Kho-Kho and secured the runners-up position.

CAT 2023 – 93.28 percentile

Achieved 93.28 percentile in the Common Admission Test (CAT) 2023, which assessed Quantitative Aptitude, Data Interpretation & Logical Reasoning, and Verbal Ability & Reading Comprehension.

Declaration : I hereby declare that the information provided in this resume is true and correct to the best of my knowledge.

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